Course notes

The coordinate calculations in this chapter are done in an orthonormal coordinate system of space.

Scalar product in space

For non-zero vectors $\vec{u}, \vec{v}$, $\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\| \cos\theta$ with $\theta \in [0, \pi]$ is their angle. If one is zero, the scalar product is 0.

$$\vec{u} \cdot \vec{v} = \tfrac{1}{2}(\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2).$$

In an orthonormal coordinate system, for $\vec{u} = (x, y, z)$ and $\vec{v} = (x', y', z')$,

$$\vec{u} \cdot \vec{v} = xx' + yy' + zz'.$$

The scalar product is symmetric and bilinear; $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$. Two vectors are orthogonal if and only if their scalar product is zero.

Example. $\vec{u} = (1, 2, 2)$ and $\vec{v} = (2, -1, 0)$ are orthogonal: $2 \quad -2 + 0 = 0$. Their norms are 3 and $\overline{5}$.

Line, plane and normal vector

Two lines are orthogonal if their direction vectors are. A line is orthogonal to a plane if it is orthogonal to two non-collinear directions of this plane. A vector normal to a plane is a non-zero vector orthogonal to all its directions.

Why two directions are enough. If $\vec{n} \cdot \vec{u} = \vec{n} \cdot \vec{v} = 0$, then, for every $\vec{w} = s\vec{u} + t\vec{v}$, $\vec{n} \cdot \vec{w} = s\vec{n} \cdot \vec{u} + t\vec{n} \cdot \vec{v} = 0$.

Vector condition for membership

If A lies in the plane $\mathcal{P}$ with normal vector $\vec{n}$, then $M \in \mathcal{P}$ if and only if $\overrightarrow{AM} \cdot \vec{n} = 0$.

Orthogonal projection and distance

The orthogonal projection H of M onto a plane is the unique point of the plane such that $\overrightarrow{MH}$ is normal to the plane ; if M lies in the plane, $H = M$. The distance from M to the plane is MH. On a line with direction vector $\vec{u}$, the projection satisfies $H \in d$ and $\overrightarrow{MH} \cdot \vec{u} = 0$.

Proof of the minimum. For every point P of the plane, $\overrightarrow{MH} \perp \overrightarrow{HP}$, therefore $MP^2 = MH^2 + HP^2 \geq MH^2$. Equality holds only for $P = H$. The projection is the point of the plane closest to it.

Method on a line. If $d = A + \mathbb{R}\vec{u}$, write $H = A + t\vec{u}$ and solve $(A - M + t\vec{u}) \cdot \vec{u} = 0$. Thus $t = \frac{\overrightarrow{AM} \cdot \vec{u}}{\|\vec{u}\|^2}$.

Angles, areas and volumes

For an angle $\widehat{ABC}$, use $\overrightarrow{BA}$ and $\overrightarrow{BC}$: $\cos \widehat{ABC} = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{BA \times BC}$. For the acute angle between two lines, take the absolute value of the quotient. The area of a triangle is $\frac{bh}{2}$; the volume of a tetrahedron is $\frac{\mathcal{A}_{\text{base}}\, h}{3}$, & h is an orthogonal distance. Units are squared and cubed respectively.

Expansions and loci

By bilinearity,

$$\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2, \qquad \|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2.$$

We can therefore also calculate $\vec{u} \cdot \vec{v} = \frac{1}{2}(\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$. For two distinct points A, B w/ midpoint I, the locus of points M such that $MA = MB$ is the plane passing through I w/ normal $\overrightarrow{AB}$: the *plan* bisector . Indeed, writing the vectors from I, expanding gives $MA^2 - MB^2 = 2\overrightarrow{IM} \cdot \overrightarrow{AB}$. Equality of distances is therefore equivalent to $\overrightarrow{IM} \cdot \overrightarrow{AB} = 0$. **Extension, with no dedicated assessment in 2026-27.** Two planes are perpendicular when their normal vectors are orthogonal. This property must not be confused with parallel planes, which correspond to collinear normals.

See a projection, then calculate it

11.1 A scalar product may be negative

First use: S48.

Definition · Scalar product

For two non-zero vectors $\vec{u}, \vec{v}$, $\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos\theta$ with $\theta \in [0; \pi]$ is their angle. In a orthonormal coordinate system, this product is $xx' + yy' + zz'$. A zero value characterises the orthogonality of directions.

Worked example · Recognising an obtuse angle

For $\vec{u} = (1;0;0)$ and $\vec{v} = (-1;1;0)$, the product is -1, the norms are 1 and $\sqrt{2}$, therefore $\cos\theta = -1/\sqrt{2}$ and $\theta = 3\pi/4$. A negative product indicates an obtuse or straight angle, not a negative length.

A1 · Immediate application Calculate the angle between $(1;1;0)$ and $(1;0;1)$ in an orthonormal coordinate system. Could the same calculation be used in any basis?

11.2 Two directions are needed

First use: S51.

Property · Criterion for orthogonality to a plane

A non-zero vector is normal to a plane if it is orthogonal to two non-collinear direction vectors of this plane. By bilinearity, it is then orthogonal to any combination of these two vectors, hence to every direction in the plane.

Worked example · Testing the three conditions

For $\vec{u} = (1;1;0)$, $\vec{v} = (0;1;1)$ and $\vec{n} = (1;-1;1)$, $\vec{n} \cdot \vec{u} = 0$ and $\vec{n} \cdot \vec{v} = 0$. The two directions are not collinear and $\vec{n} \neq 0$, therefore $\vec{n}$ is normal to the plane they span.

A2 · Counterexample "Being orthogonal to one line in the plane is enough to be normal to the plane." Disprove using two perpendicular directions in the horizontal plane. Also explain the difference between orthogonal lines and perpendicular lines.

Worked example · A locus of equidistant points

Let A=(-1;0;0) and B=(1;0;0). Their midpoint is O. For M=(x;y;z), $MA^2 - MB^2 = (x+1)^2 - (x-1)^2 = 4x$. Thus $MA = MB$ is equivalent to x=0: the locus is the plane through O, normal to $\overrightarrow{AB}$. It is not just the perpendicular bisector drawn on a single face.

A3 · Application and justification For A=(0;0;0), B=(2;2;0), describe geometrically the locus of the points equidistant from A and B. Use the squares of the distances to give a relation between the coordinates; find two points of this locus.

11.3 The foot is the nearest point

First use: S54 and S55.

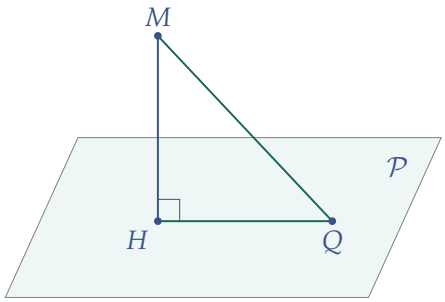


Figure 11.1. The marking shows $MH \perp \mathcal{P}$. For each Q of the plane, $MQ^2 = MH^2 + HQ^2$.

Proof. If H is the orthogonal projection of M onto a plane $\mathcal{P}$, for every $Q \in \mathcal{P}$, the triangle MHQ is right-angled at H. Thus $MQ^2 = MH^2 + HQ^2 \geq MH^2$. Equality requires $HQ = 0$, therefore $Q = H$. The minimum is reached at a single point. The reasoning is the same for a line. □

Worked example · Projecting onto a line without a plane equation

Let $A = (1;0;0)$, $M = (0;2;3)$ and a line through point A with direction $\vec{u} = (1;1;0)$. Write $\overrightarrow{AH} = t\vec{u}$ gives $H = (1+t;t;0)$. The condition $\overrightarrow{MH} \cdot \vec{u} = 0$ becomes $(1+t) + (t-2) = 0$, therefore $t = 1/2$. Thus $H = (3/2;1/2;0)$ and $MH^2 = 27/2$. We check both membership of the line and orthogonality.

A4 · Partial guidance Project $M = (2;3;1)$ onto the line through O with direction $(1;1;0)$. Determine the projection and the distance.

A5 · Choice of strategy Project $M = (1;2;3)$ onto the plane spanned by $\vec{i}, \vec{j}$. Justify geometrically the coordinates of the foot, then prove that every other point of the plane is further away.

11.4 One quantity by two calculations

First use: S56.

Worked example · Recovering an altitude

In an orthonormal coordinate system, $O = (0;0;0)$, $A = (6;0;0)$, $B = (0;4;0)$, $C = (0;0;3)$. The triangle OAB is right-angled with area 12 and OC is perpendicular to its plane; the volume is 12 $\times 3/3 = 12$. If the area of ABC is 3 $\overline{29}$, its altitude from O is 3 $V/\mathcal{A}(ABC) = 12/\ \overline{29}$.

A6 · Synthesis In this tetrahedron, calculate AB, AC and $\overrightarrow{AB} \cdot \overrightarrow{AC}$. Deduce the cosine and then the sine of the angle $\widehat{BAC}$, check the area of ABC and recover the altitude from O.

A first step: Use $\mathcal{A}(ABC) = \frac{1}{2}$AB AC sin $\widehat{BAC}$. An angle of a triangle has a positive sine.

A7 · Fix a check A candidate gives a point H of the plane and states "it is the projection since H lies in the plane". What second check is missing? Give two points of the plane $z = 0$ to show that membership alone is not enough.

One minute, spoken aloud. Describe the projection figure, naming the external point, the plane, the foot and the right angle; explain why we do not simply calculate the distance to any point of the plane.

Delayed review · S70. Recalculate the foot with a parametric representation and a Cartesian equation; the two geometric conditions remain the same.

Investigation

The nearest point on a line - 35 min.

In an orthonormal coordinate system of space, consider $A(1;0;0)$, $M(0;2;3)$ and the line d through A, with direction vector $\vec{u} = (1;1;0)$. We look for the point on d closest to M.

1. Express, in terms of a real parameter t, the coordinates of a point H of d.
2. Determine the value of t for which $\overrightarrow{MH} \cdot \vec{u} = 0$. Deduce the coordinates of the point H thus obtained.
3. Calculate distance MH.
4. Let P be any point of d. Prove that $MP^2 = MH^2 + HP^2$, then deduce that H is the unique point on d closest to M.